

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
The automobile battery produces 12 volts and the head light bulb dissipates 40 watts of joule heat . We know the formula for power (P) in terms of voltage (V) and current (I) is $P = V * I$ and in terms of voltage and resistance (R) is $P = V^2 / R$. We will first rearrange the power formula in terms of current to find the current (I) using the known powers and voltage values . $P = V * I$ can be rearranged to $I = P / V$. Substituting the power and voltage , the current will then be $I = 40 / 12$ amps . Next , we will use this value for calculating the resistance (R) eskorte The </think><answer>3 . 33 amps</answer>